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CUSTOMER	: IDAMAN PHARMA MANUFACTURING SDN BHD
DESCRIPTION	: Levozal 5mg
MEASUREMENT	: W:148.5mm X H:210mm
MATERIAL	: SPMO 050

COLOUR : BLACK

FINISHING : FOLD

MATERIAL CODE:	Attn : <u>MS ZAKIAH</u>
Lt-XXX.XX	From : <u>MAJALA</u> I hereby verify & confirm that this draft is the final copy. Any subsequent changes of the design and content of the end product will not be borne by THUNDER PRINT SDN. BHD.
	Customer Signature & Date

14.6.2022



PRODUCT DESCRIPTION

White to off white, oval, film-coated tablet with a 'LS' logo on one side and plain on the other.

Each film-coated tablet contains 5mg of Levocetirizine Dihydrochloride.

ROUTE OF ADMINISTRATION

For Oral Use.

PHARMACODYNAMICS

Pharmacotherapeutic group: antihistamine for systemic use, piperazine derivatives. ATC code: R06A E09.

Mechanism of action

Levocetirizine, the (R) enantiomer of cetirizine, is a potent and selective antagonist of peripheral H_1 -receptors.

Binding studies revealed that levocetirizine has high affinity for human H_1 -receptors ($K_i = 3.2$ nmol/l). Levocetirizine has an affinity 2-fold higher than that of cetirizine ($K_i = 6.3$ nmol/l). Levocetirizine dissociates from H_1 -receptors with a half-life of 115 ± 38 min.

After single administration, levocetirizine shows a receptor occupancy of 90% at 4 hours and 57% at 24 hours.

Published data of pharmacodynamic studies in healthy volunteers demonstrate that, at half the dose, levocetirizine has comparable activity to cetirizine, both in the skin and in the nose.

Available published data comparing the effects of levocetirizine 5mg, desloratadine 5mg, and placebo on histamine-induced wheal and flare, indicates that levocetirizine treatment resulted in significantly decreased wheal and flare formation which was highest in the first 12 hours and lasted for 24 hours, ($p < 0.001$) compared with placebo and desloratadine.

The onset of action of levocetirizine 5 mg in controlling pollen-induced symptoms has been observed at 1 hour post drug intake in placebo controlled trials in the model of the allergen challenge chamber.

Available data in vivo studies (Boyden chambers and cell layers techniques) show that levocetirizine inhibits cotaxin-induced eosinophil transendothelial migration through both dermal and lung cells. Published data from a pharmacodynamic experimental study in vivo (skin chamber technique) showed three main inhibitory effects of levocetirizine 5 mg in the first 6 hours of pollen-induced reaction, compared with placebo in 14 adult patients: inhibition of VCAM-1 release, modulation of vascular permeability and a decrease in eosinophil recruitment.

PHARMACOKINETICS

The pharmacokinetics of levocetirizine are linear with dose- and time- independent with low inter-subject variability. The pharmacokinetic profile is the same when given as the single enantiomer or when given as cetirizine. No chiral inversion occurs during the process of absorption and elimination.

Absorption

Levocetirizine is rapidly and extensively absorbed following oral administration. In adults, peak plasma concentrations are achieved 0.9 hour after dosing. Steady state is achieved after two days. Peak concentrations are typically 270 ng/ml and 308 ng/ml following a single and a repeated 5 mg once daily dose, respectively. The extent of absorption is dose independent and is not altered by food, but the peak concentration is reduced and delayed.

Distribution

No tissue distribution data is available in humans, neither concerning the passage of levocetirizine through the blood-brain barrier. In rats and dogs, the highest tissue levels, are found in liver and kidneys, the lowest in the central nervous system (CNS) compartment. In humans, levocetirizine is 90% bound to plasma proteins. The distributions of levocetirizine is restrictive, as the volume of distribution is 0.4 l/kg.

Metabolism

The extent of metabolism of levocetirizine in humans is less than 14% of the dose and therefore differences resulting from genetic polymorphism or concomitant intake of enzyme inhibitors are expected to be negligible. Metabolic pathways include aromatic oxidation, N- and O-dealkylation and taurine conjugation. Dealkylation pathways are primarily mediated by CYP3A4 while aromatic oxidation involved multiple and/or unidentified CYP isoforms. Levocetirizine had no effect on the activities of CYP isoenzymes 1A2, 2C9, 2C19, 2D6, 2E1 and 3A4 at concentrations well above peak concentrations achieved following a 5 mg oral dose.

Due to its low metabolism and absence of metabolic inhibition potential, the interaction of levocetirizine with others substances, or vice-versa, is unlikely.

Elimination

The plasma half-life in adults is 7.9 ± 1.9 hr. The half-life is shorter in small children. The mean apparent total body clearance in adults is 0.63 ml/min/kg. The major route of excretion of levocetirizine and metabolites is via urine, accounting for a mean of 85.4% of the dose. Excretion via faeces accounts for only 12.9% of the dose. Levocetirizine is excreted both by glomerular filtration and active tubular secretion.

Special patient populations

Children

Data from a paediatric pharmacokinetic study with oral administration of a single dose of 5 mg levocetirizine in 14 children age 6 to 11 years with body weight ranging between 20 and 40 kg show that C_{max} and AUC values are about 2-fold greater than that reported in healthy adult subjects in a cross-study comparison. The mean C_{max} was 450 ng/mL, occurring at a mean time of 1.2 hours, weight-normalised, total body clearance was 30% greater, and the elimination half-life 24% shorter in this paediatric population than in adults.

Elderly

Limited pharmacokinetic data are available in elderly subjects.

Following once daily repeat oral administration of 30 mg levocetirizine for 6 days in 9 elderly subjects (65–74 years of age), the total body clearance was approximately 33% lower compared to that in younger adults. The disposition of racemic cetirizine has been shown to be dependent on renal function rather than on age. This finding would also be applicable for levocetirizine, as levocetirizine and cetirizine are both predominantly excreted in urine. Therefore, the levocetirizine dose should be adjusted in accordance with renal function in elderly patients.

Renal impairment

The apparent body clearance of levocetirizine is correlated to the creatinine clearance. It is therefore recommended to adjust the dosing intervals of levocetirizine, based on creatinine clearance in patients with moderate and severe renal impairment. In anuric end stage renal disease subjects, the total body clearance is decreased by approximately 80% when compared to normal subjects. The amount of levocetirizine removed during a standard 4 hours haemodialysis procedure was < 10%.

Hepatic impairment

The pharmacokinetics of levocetirizine in hepatically impaired subjects have not been tested. Patients with chronic liver diseases (hepatocellular, cholestatic, and biliary cirrhosis) given 10 or 20 mg of the racemic compound cetirizine as a single dose had a 50% increase in half-life along with a 40% decrease in clearance compared to healthy subjects.

Other patient characteristics

Gender

Pharmacokinetic results for 77 patients (40 men, 37 women) were evaluated for potential effect of gender. The half-life was slightly shorter in women (7.08 ± 1.72 hr) than in men (8.62 ± 1.84 hr); however, the body weight-adjusted oral clearance in women (0.67 ± 0.16 mL/min/kg) appears to be comparable to that in men (0.59 ± 0.12 mL/min/kg). The same daily doses and dosing intervals are applicable for men and women with normal renal function.

Race

The effect of race on levocetirizine has not been studied. As levocetirizine is primarily renally excreted, and there are no important racial differences in creatinine clearance, pharmacokinetic characteristics of levocetirizine are not expected to be different across races. No race-related differences in the kinetics of racemic cetirizine have been observed.

INDICATIONS

Levocetirizine is indicated for the symptomatic treatment of allergic rhinitis (including persistent allergic rhinitis) and chronic idiopathic urticaria.

RECOMMENDED DOSAGE

The film-coated tablet must be taken orally, swallowed whole with liquid and may be taken with or without food. It is recommended to take the daily dose in one single intake.

Adults and adolescents 12 years and above:

The daily recommended dose is 5 mg (1 film-coated tablet) once daily.

Children aged 6 to 12 years:

The daily recommended dose is 5 mg (1 film-coated tablet) once daily.

Elderly:

Adjustment of the dose is recommended in elderly patients with moderate to severe renal impairment.

Renal impairment:

The dosing intervals must be individualized according to renal function. Refer to the following table and adjust the dose as indicated. To use the dosing table, an estimate of the patient's creatinine clearance (CL_{CR}) in mL/min is needed. The CL_{CR} (mL/min) may be estimated from serum creatinine (mg/dl) determination using the following formula:

$$CL_{CR} = \left[\frac{140 - \text{age (years)}}{72} \right] \times \frac{\text{weight (kg)}}{\text{serum creatinine (mg/dl)}} \times 0.85 \text{ (for women)}$$

Dosing Adjustments for Patients with Impaired Renal Function:

Group	Creatinine Clearance (mL/min)	Dosage and Frequency
Normal	≥80	5 mg once daily
Mild	50 – 79	5 mg once daily
Moderate	30 – 49	5 mg once daily once every 2 days
Severe	<30	5 mg once daily once every 3 days
End-stage renal disease - Patients undergoing dialysis	<10	Contraindicated

In paediatric patients suffering from renal impairment, the dose will have to be adjusted on an individual basis taking into account the renal clearance of the patient and his body weight. There is no specific data for children with renal impairment.

Hepatic impairment:

No dose adjustment is needed in patients with solely hepatic impairment. In patients with hepatic impairment and renal impairment, adjustment of the dose is recommended.

Duration of use

Intermittent allergic rhinitis (symptoms < 4 days/ week or less than 4 weeks a year) has to be treated according to the disease and its history; it can be stopped once the symptoms have disappeared and can be restarted again when symptom reappear.

In case of persistent allergic rhinitis (symptoms > 4 days/week and for more than 4 weeks a year), continuous therapy can be proposed to the patient during the period of exposure to allergens.

Clinical experience with 5mg levocetirizine as a film-coated tablet formulation is currently available for a 6-months treatment period. The chronic urticaria and chronic allergic rhinitis, up to one year's clinical experience is available for the racemate.

CONTRAINDICATIONS

Levocetirizine is contraindicated in:

- hypersensitivity to levocetirizine, to cetirizine, to hydroxyzine, to any piperazine derivatives or any of the excipients contained in the product.
- severe renal impairment at less than 10 ml/min creatinine clearance.
- rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption.

WARNINGS AND PRECAUTIONS

Do not exceed the stated dose.

Alcohol

Precaution is recommended with intake of alcohol.

Risk of urinary retention

Caution should be taken in patients with predisposing factors of urinary retention (e.g. spinal cord lesion, prostatic hyperplasia) as levocetirizine may increase the risk of urinary retention.

Risk of seizure aggravation

Caution should be taken in patients with epilepsy and patients at risk of convulsion as levocetirizine may cause seizure aggravation.

Allergy skin tests

Response to allergy skin tests are inhibited by antihistamines and a wash-out period (of 3 days) is required before performing them.

Withdrawal syndrome

Pruritus may occur when levocetirizine is stopped even if those symptoms were not present before treatment initiation. The symptoms may resolve spontaneously. In some cases, the symptoms may be intense and may require treatment to be restarted. The symptoms should resolve when the treatment is restarted.

Excipient: Lactose

This medical product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

INTERACTIONS WITH OTHER MEDICAMENTS

No interaction studies have been performed with levocetirizine (including no studies with CYP3A4 inducers); studies with the racemate compound cetirizine demonstrated that there were no clinically relevant adverse interactions antipyrine (with antipyrine pseudoephedrine, cimetidine, ketoconazole, erythromycin, azithromycin, glipizide and diazepam).

Theophylline

A small decrease in the clearance of cetirizine (16%) was observed in a multiple dose study with theophylline (400 mg once a day); while the disposition of theophylline was not altered by concomitant cetirizine administration.

Ritonavir

In a multiple dose study of ritonavir (600 mg twice daily) and cetirizine (10 mg) daily, the extent of exposure to cetirizine was increased by about 40% while the disposition of ritonavir was slightly altered (-11%) further to concomitant cetirizine administration.

Food

The extent of absorption of levocetirizine is not reduced with food, although the rate of absorption is decreased.

Alcohol

In sensitive patients, the concurrent administration of cetirizine or levocetirizine and alcohol or other CNS depressants may cause additional reductions in alertness and impairment of performances.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

No evidence that levocetirizine at the recommended dose impairs mental alertness, reactivity or the ability to drive. Nevertheless, some patients could experience somnolence, fatigue and asthenia under therapy with levocetirizine. Therefore, patients intending to drive, engage in potentially hazardous activities or operate machine- ery should take their response to the medicinal product into account.

PREGNANCY AND LACTATION

Pregnancy

The use of levocetirizine may be considered during pregnancy, if necessary.

Lactation

Caution should be exercised when prescribing to lactating women. Cetirizine,the racemate of levocetirizine has been shown excreted in human. Therefore, the excretion of levocetirizine in human milk is likely. Adverse reactions associated with levocetirizine may be observed in breastfed infants.

SIDE/ADVERSE EFFECTS

Nervous system disorders

Common: headache, somnolence

Gastrointestinal disorders

Common: dry mouth

Uncommon: abdominal pain

General disorders and administration site conditions

Common: fatigue

Uncommon: asthenia

Immune system disorders

Not known: hypersensitivity including anaphylaxis

Metabolism and nutrition disorders

Not known:increase weight, increase appetite

Psychiatric disorders

Not known: aggression, agitation, hallucination, depression, insomnia, suicidal ideation, nightmares.

Nervous system disorders

Not known: convulsions, paraesthesia, dizziness, syncope, tremor, dysgeusia

Eye disorders

Not known:visual disturbances, blurred vision, oculogyration

Ear and labyrinth disorders

Not known: vertigo

Cardiac disorders

Not known:palpitations, tachycardia

Respiratory, thoracic, and mediastinal disorders

Not known:dyspnoea

Gastrointestinal disorders

Not known: nausea, vomiting, diarrhoea

Hepatobiliary disorders

Not known: hepatitis, abnormal liver function test

Skin and subcutaneous tissue disorders

Not known: angioneurotic oedema, fixed drug eruption, pruritus, rash, urticaria

After levocetirizine discontinuation, pruritus has been reported.

Musculoskeletal and connective tissue disorders

Not known: myalgia, arthralgia

Renal and urinary disorders

Not known: dysuria, urinary retention

General disorders and administration site conditions

Not known:oedema

SYMPTOMS& TREATMENT FOR OVERDOSAGE

Symptoms and Signs

Symptoms observed after an overdose of levocetirizine are mainly associated with CNS effects or with effects that could suggest an anticholinergic effect.

Symptoms of overdose may include drowsiness in adults and initially agitation and restlessness, followed by drowsiness in children. Adverse events reported after an intake of at least 5 times the recommended daily dose are: confusion, diarrhea, dizziness, fatigue, headache, malaise, mydriasis, pruritus, restlessness, sedation, somnolence, stupor, tachycardia, tremor and urinary retention.

Treatment

There is no known specific antidote to levocetirizine. Should overdose occur, symptomatic or supportive treatment is recommended. Gastric lavage should be considered following ingestion of a short occurrence. Levocetirizine is not effectively removed by haemodialysis.

STORAGE CONDITION

Store below 30°C. Do not use after the expiry date stated on the unit carton and blister.

SHELF LIFE

2 years from the date of manufacture.

PACK SIZE

Blister pack of 10's

i) 10 strips of 10's

ii) 50 strips of 10's

REGISTRATION NUMBER : MALXXXXXXXXXXXXXXXXX

CONTROLLED MEDICINE

UBAT TERKAWAL

KEEP MEDICINES OUT OF REACH OF CHILDREN

JAUHI UBAT-UBATAN DARI KANAK-KANAK

For further information, please consult your doctor or your pharmacist.

Revision Number : 00

Revision Date : 14 June 2022

Lt-XXX.XX PRODUCT REGISTRATION HOLDER AND MANUFACTURER

IDAMAN PHARMA MANUFACTURING SDN BHD (20040102395)

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